

GLOBAL E-COMMERCE SALES & PROFITABILITY DASHBOARD

End-to-End Data Analytics Project

Using

Python | Power BI | DAX | Data Modeling

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PROJECT OVERVIEW

Introduction

The **Global E-Commerce Sales & Profitability Dashboard** is an end-to-end Data Analytics project developed to analyze worldwide commercial transactions using Python, Power BI Desktop, DAX, and relational data modeling.

The project focuses on transforming raw transactional datasets into interactive, multi-dimensional executive dashboards that track financial performance, supply chain shipping logistics, and product profit margins.

Project Purpose

The core objective of this project is to assist executive leadership and operations managers in identifying high-margin product lines, uncovering logistics cost leakages, evaluating customer purchasing power across segments, and monitoring macro KPIs to guide strategic planning.

Project Workflow



PROJECT OBJECTIVES

Key Objectives

- Analyze overall commercial sales performance (\$485K+ revenue and 32.8% profit margin).
- Track year-over-year revenue trends, seasonal velocity, and monthly sales spikes.
- Analyze profitability across major customer segments (Consumer, Corporate, Home Office).
- Measure the impact of freight and shipping logistics expenses on net operational margins.
- Evaluate customer payment method adoption (Credit Card, Net Banking, COD).
- Examine product categories and sub-categories to isolate top revenue drivers.
- Identify loss-making products and evaluate price-vs-quantity demand sensitivity.
- Design an interactive 3-page reporting layout in Power BI with dynamic page slicers.
- Develop robust DAX formulas for dynamic KPI calculation across cross-filtered dimensions.
- Deliver executive-level recommendations to eliminate margin leakage and boost bottom line.

Expected Outcome

This project provides a single source of truth for business performance. It replaces manual spreadsheet tracking with a live BI system that empowers decision-makers to quickly evaluate financial health, audit fulfillment expenses, and capitalize on high-margin product opportunities.

DATASET DESCRIPTION

Dataset Overview

This project utilizes a multi-regional E-Commerce Transactional dataset comprising international orders across diverse consumer products. It records order details, product attributes, financial metrics (revenue, discount, unit price), freight charges, and customer demographic segments.

Property	Value
Dataset Name	Global E-Commerce Sales Dataset
File Format	CSV (Comma-Separated Values)
Total Records	4,950+ Orders (7,420 Units Sold)
Exploration Tool	Python (Pandas, NumPy)
Visualization Tool	Power BI Desktop
Calculated Measures	DAX (Data Analysis Expressions)
Target Metrics	Sales, Profit, Profit Margin %, Shipping Cost

Important Columns & Attributes

- **Order ID & Order Date:** Unique order identifier and transaction date for time-intelligence analysis.
- **Customer Name & Segment:** Purchaser classification (Consumer, Corporate, Home Office).
- **Country, City & Region:** Geographic location attributes for logistics mapping.
- **Category & Sub-Category:** Product classifications (Technology, Furniture, Office Supplies).
- **Product Name:** Granular individual item description.
- **Sales, Quantity & Profit:** Core monetary revenue, volume, and net financial earnings.
- **Shipping Cost & Payment Method:** Fulfillment charges and payment gateway classification.

PYTHON DATA EXPLORATION

Data Cleaning & Preprocessing

Before connecting the dataset to Power BI, Python scripts (ecommerce_analysis.py) were used to conduct exploratory data analysis (EDA), detect missing attributes, validate schema data types, and ensure integrity.

Implementation Steps in Python

- Loaded raw transaction files into Pandas DataFrames for schema inspection.
- Identified and addressed null and outlier values across pricing and margin columns.
- Formatted temporal fields into standardized datetime format for Power BI date hierarchies.
- Calculated baseline correlation matrices between discount levels, order quantity, and gross profit.
- Exported clean, verified data (global_ecommerce_sales.csv) ready for Power BI consumption.

Core Python Code Excerpt

```
import pandas as pd
df = pd.read_csv('global_ecommerce_sales.csv')
df['Order Date'] = pd.to_datetime(df['Order Date'])
df['Profit Margin'] = df['Profit'] / df['Sales']
print(df.groupby('Category')[['Sales', 'Profit']].sum())
```

Key Python Learning

This preprocessing stage validated that the dataset had zero corrupt transactions, ensured high integrity for Power BI import, and reinforced scripting skills in automated data cleansing and preliminary descriptive analytics.

DATA MODELING & ARCHITECTURE

Relational Schema Design

To optimize analytical performance and enable seamless cross-filtering across three distinct dashboard pages, a clean star-schema design was implemented in Power BI Desktop.

Modeling Implementation Steps

- Organized transactional records into a central **Fact_Orders** table containing granular metrics.
- Created Dimension tables: **Dim_Customers**, **Dim_Products**, **Dim_Geography**, and **Dim_Calendar**.
- Established 1-to-Many (1:*) active relationships between dimensions and the fact table.
- Configured single-directional cross-filtering to prevent circular dependencies and ambiguous pathways.
- Configured an independent Measures Table to house all custom DAX KPIs cleanly.

Model Benefits

- **High Query Speed:** Instant visual rendering across 4,950+ transaction rows.
- **Cross-Filtering Integrity:** Selecting any visual dynamically recalibrates all related metrics across the canvas.
- **Scalability:** Enables smooth integration of future ERP or SQL database connectors without rebuilding visual logic.

POWER BI: EXECUTIVE SUMMARY

Dashboard Overview: Page 1

The first dashboard view delivers high-level visibility for C-suite and commercial directors, providing an immediate snapshot of top-line revenue, net profitability, customer segment breakdown, and multi-year trajectory.

Dashboard Components & Visuals

- **Executive KPI Cards:** Displays Total Sales (\$485K), Net Profit (\$159K), Profit Margin (32.8%), and Total Units (7,420).
- **Yearly Sales & Profit Trend:** Multi-line chart illustrating continuous revenue expansion and seasonal peaks.
- **Customer Segment Breakdown:** Donut chart displaying Consumer segment generating ~51% of total earnings.
- **Category Revenue Contribution:** Horizontal bar chart ranking Technology, Office Supplies, and Furniture.
- **Interactive Filters:** Dynamic date slicers and regional drop-downs allowing instant drilldown.

Business Value Delivered

Enables senior leadership to track core business pulse within seconds of loading the report, verifying that healthy profit margins (32.8%) are maintained across operating quarters.

LOGISTICS & CUSTOMER INSIGHTS

Dashboard Overview: Page 2

The second page focuses on supply chain fulfillment, shipping cost dynamics, customer acquisition tiers, and regional market distribution across domestic and international trade zones.

Dashboard Components & Visuals

- **Country & Regional Distribution:** Map and bar charts mapping revenue intensity across North America, Europe, APAC, and LATAM.
- **Shipping Cost vs. Margin Impact:** Clustered analysis comparing shipping freight expenses to order volume to detect margin erosion.
- **Payment Mode Breakdown:** Share analysis identifying Credit Cards and Net Banking as driving >75% of cash flow.
- **Top Customer Contribution:** Ranked bar chart listing top repeat purchasers and their cumulative lifetime value.
- **Delivery Speed Analysis:** Fulfillment lead-time tracking across standard, second-class, and express shipping.

Operational Value

Assists logistics teams in renegotiating carrier rates in high-cost remote zones and guides marketing teams toward rewarding high-frequency enterprise customers.

PRODUCT PROFITABILITY ANALYSIS

Dashboard Overview: Page 3

The third view provides granular drill-down into product inventory, isolating which items yield high returns and which lines suffer from margin compression or loss-leader discounting.

Dashboard Components & Visuals

- **Product Profitability Matrix:** Hierarchical table displaying Product Name, Category, Unit Price, Units Sold, Total Sales, and Net Margin %.
- **Price vs. Quantity Elasticity:** Scatter matrix comparing retail unit prices against demand volumes to detect pricing sensitivity.
- **Top & Bottom Performers:** Filtered ranking identifying top profit generators (e.g. Ergonomic Office Chairs, LED Monitors).
- **Discount Impact Visual:** Evaluates whether heavy promotional discounts (>20%) successfully generated compensatory sales volumes.
- **Dynamic Product Slicer:** Allows interactive filtering by category, price band, and profitability threshold.

Merchandising Value

Supplies merchandising and procurement heads with concrete data to eliminate negative-margin items, optimize inventory turnover, and adjust promotional discount guardrails.

DAX MEASURES & CALCULATIONS

Introduction to DAX

Data Analysis Expressions (DAX) were authored in Power BI to construct calculated measures. These measures compute aggregated metrics dynamically based on report slicers and interactive visual cross-filtering.

Core DAX Formulations

Total Sales

Total Sales = SUM(Orders[Sales])

Calculates total gross transaction revenue across all filtered orders.

Total Profit

Total Profit = SUM(Orders[Profit])

Calculates total net profit generated after product cost deduction.

Profit Margin %

Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)

Computes robust percentage profit margin, safely handling zero denominators.

Average Discount

Average Discount = AVERAGE(Orders[Discount])

Tracks mean promotional discount applied across customer orders.

Total Units Sold

Total Units = SUM(Orders[Quantity])

Computes total volume of products fulfilled across all channels.

STRATEGIC BUSINESS INSIGHTS

Key Data Discoveries

- **Technology & Office Supplies Lead Profitability:** Technology delivers 35.0% profit margin and Office Supplies delivers 34.0%, while Furniture trails at 28.0% due to bulk logistics overhead.
- **Consumer Segment Accounts for Half of Sales:** Consumer orders generate 51% (\$247K+) of revenue with steady 32.8% margin consistency.
- **Shipping Cost Leakages in Remote Zones:** Freight expenses in certain secondary logistics corridors erode up to 15% of gross order margin on low-ticket baskets.
- **Ineffective Steep Discounting:** Price reductions exceeding 20% failed to trigger proportional unit volume growth in furniture categories.
- **Digital Payment Dominance:** Over 75% of gross revenue is processed through Credit Card and Electronic Net Banking, providing rapid cash settlement.

Strategic Actionable Recommendations

- **Implement Free Shipping Threshold:** Enforce a \$50 minimum order threshold for free delivery to protect margin on low-value orders.
- **Establish Discount Caps:** Restrict discretionary promotional discounts to 15% maximum, shifting to bundled accessory cross-selling.
- **Strengthen Vendor Supply Lines:** Prioritize top-performing SKUs (Ergonomic Chairs, Monitors) to ensure zero stockout in peak Q4 demand.

SKILLS APPLIED & TOOLS

Technical Competencies Demonstrated

This project showcases a complete technical skillset across data preparation, business intelligence modeling, and executive storytelling:

Domain Area	Tools & Techniques
Programming & Exploration	Python, Pandas, NumPy, Data Cleaning, EDA
Business Intelligence	Power BI Desktop, Multi-page UI/UX Design, Slicers
Data Modeling	Star Schema, 1-to-Many Relationships, Fact/Dim Tables
Calculated Measures	DAX (SUM, DIVIDE, AVERAGE, Time-Intelligence)
Logistics & Analytics	Freight Cost Analysis, Elasticity Matrix, Drill-downs
Version Control & Portfolio	Git, GitHub Repo, HTML/CSS Web Integration

Methodological Rigor

By combining exploratory Python scripting with an interactive Power BI architecture, this project reflects enterprise-grade analytical rigor suited for Data Analyst, Business Intelligence, and Reporting roles.

CONCLUSION & FUTURE SCOPE

Project Summary

The **Global E-Commerce Sales & Profitability Dashboard** successfully demonstrates a comprehensive business intelligence pipeline. It transforms raw transactional rows into structured intelligence, providing granular clarity on revenue drivers, logistics margins, and customer demand patterns.

Key Project Achievements

- Cleaned and preprocessed 4,950+ international orders using Python scripts.
- Modeled a robust star-schema relational model in Power BI Desktop.
- Engineered custom DAX measures for real-time KPI dynamic filtering.
- Built a 3-page interactive dashboard covering Executive, Logistics, and Product views.
- Identified actionable margin optimization pathways to curb logistics leakage.
- Published full project documentation and repository on GitHub and portfolio website.

Future Enhancements

- **Machine Learning Forecasting:** Integrate predictive time-series models for future revenue projections.
- **Automated Direct Pipeline:** Establish live database connectivity via SQL/PostgreSQL for auto-refresh.
- **RFM Segmentation:** Build Recency-Frequency-Monetary customer clustering models.

Final Statement

This project highlights my ability to deliver end-to-end data analytics solutions that translate complex data into clear, strategic business value for stakeholders.